

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

PROJECT TITLE: DESIGN AND DEVELOPMENT OF AN INFORMATIVE WEBSITE FOR THE DANGOTE EAST AFRICA PETROLEUM REFINERY PROJECT IN KENYA

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1. INTRODUCTION

1.1 Purpose of the Document

The purpose of this Software Requirements Specification (SRS) document is to describe the requirements of the proposed Dangote East Africa Petroleum Refinery Informative Website. It explains the functions, features, performance requirements and limitations of the system.

The document will guide the development of the website by explaining what the system is expected to do and how users will interact with it. It will also act as a reference during system development, testing and evaluation.

1.2 Scope of the System

The proposed system is an informative website designed to provide centralized information about the Dangote East Africa Petroleum Refinery Project in Kenya.

The website will bring together information that is currently available from different websites and sources. It will allow users to access information about the background of the

project, its location, facilities, expected economic benefits, employment opportunities, environmental and social aspects, and development progress.

The system will provide a simple interface that can be accessed through computers, tablets and mobile phones.

The website will also have an administrator section that will allow authorized users to manage website content and publish updates. Users will be able to submit inquiries and subscribe to notifications about important announcements.

The system is intended for information sharing and public awareness. It will not be used to operate the refinery or manage its industrial activities.

1.3 Objectives of the System

The main objective is to design and develop an informative and user-friendly website that provides information about the Dangote East Africa Petroleum Refinery Project in Kenya.

The specific objectives are:

1. To provide background information about the Dangote East Africa Petroleum Refinery Project.
2. To provide information about the location and major facilities of the refinery.
3. To provide information about the expected economic and employment benefits of the project.
4. To provide information about the environmental and social aspects of the project.
5. To provide updates on the progress and development of the refinery.
6. To develop a responsive website that can be accessed using computers and mobile phones.
7. To provide a platform where users can submit inquiries and receive important announcements.

1.4 Definitions and Abbreviations

- SRS: Software Requirements Specification.
- Administrator: An authorized person responsible for managing website content.
- User: Any person accessing the website.
- Database: An organized collection of information stored electronically.

- SMS: Short Message Service used to send text notifications.
- Responsive Website: A website that adjusts its layout to different screen sizes.
- PHP: A server-side programming language used in web development.
- MySQL: A database management system used to store website information.
- HTML: Hypertext Markup Language used to structure web pages.
- CSS: Cascading Style Sheets used to design and style web pages.
- JavaScript: A programming language used to add interactive features to websites.

2. OVERALL SYSTEM DESCRIPTION

2.1 Product Perspective

The Dangote East Africa Petroleum Refinery Informative Website will be a web-based information system.

It will provide a central platform where users can access organized information about the refinery project without having to search through several different websites.

The system will consist of a public website, an administrator section and a database. The public website will display information to visitors, while the administrator section will allow authorized users to add, edit, update and remove website content.

The website will mainly use HTML, CSS and JavaScript for the front end. PHP and MySQL may be used to support database and administrative functions.

2.2 System Architecture

The system will operate through the following components:

1. User Interface: Allows visitors to access and interact with the website.
2. Application: Processes requests, manages website functions and communicates with the database.
3. Database: Stores website information, administrator details, inquiries and subscriber records.
4. Administrator Panel: Allows authorized administrators to manage website information.
5. SMS Gateway: Supports the sending of important notifications to users who have subscribed.

The general system flow will be:

User → Website Interface → Application → Database

Administrator → Admin Panel → Application → Database

Application → SMS Gateway → Subscribed Users

2.3 User Categories

The system will have the following categories of users:

1. General Visitors

These are members of the public who visit the website to access information about the refinery.

They will be able to:

- View information about the project.
- Read news and development updates.
- View images and other available resources.
- Search for information.
- Submit inquiries.
- Subscribe to notifications.

2. Administrator

The administrator will be responsible for managing the website.

The administrator will be able to:

- Log into the administration section.
- Add new information.
- Edit existing information.
- Publish news and announcements.
- Delete outdated or incorrect content.
- View inquiries submitted by users.
- Manage subscriber information.

- View SMS notification records.

3. Website Subscribers

These are users who provide their phone numbers to receive important announcements and updates.

They will be able to:

- Subscribe to SMS notifications.
- Receive important project announcements.
- Request to stop receiving notifications.

2.4 Operating Environment

The system will operate in a web-based environment.

The requirements include:

- A computer or mobile device.
- An internet connection.
- A modern web browser such as Google Chrome, Microsoft Edge or Mozilla Firefox.
- A web server to host the website.
- A database server for storing information.
- PHP and MySQL where database functionality is implemented.

2.5 Main System Features

The website will provide the following main features:

1. Homepage displaying an introduction to the refinery project.
2. Background information about the project.
3. Information about the location and facilities.
4. Information about expected economic and employment benefits.
5. Environmental and social information.
6. News and development updates.
7. Image gallery.

8. Search and navigation functions.
9. Public inquiry form.
10. SMS notification subscription.
11. Administrator login and content management.

2.6 User Characteristics

The system is intended for different categories of users, including students, researchers, members of the public and other people interested in the refinery project.

Users are not expected to have advanced computer skills. Therefore, the website should have simple navigation, clear information and easy-to-understand instructions.

Administrators should have basic computer skills and knowledge of website content management.

2.7 Assumptions and Dependencies

The system will operate under the following assumptions:

1. Users will have access to an internet connection.
2. Users will access the website using supported web browsers.
3. Information published on the website will be obtained from reliable sources.
4. Administrators will be responsible for updating website content.
5. SMS notifications will depend on the availability of an SMS gateway service.
6. Website hosting and database services will be available when required.
7. Users who subscribe to SMS notifications will provide valid phone numbers.

3. FUNCTIONAL REQUIREMENTS

Functional requirements describe the specific activities and services that the system must perform.

3.1 Project Information Management

FR-01: Homepage Display

The system shall display a homepage containing the project title, introduction, navigation menu and important information about the refinery.

FR-02: Project Background Information

The system shall provide information about the history and background of the Dangote East Africa Petroleum Refinery Project.

FR-03: Project Location

The system shall provide information about the location of the refinery project in Lamu, Kenya.

FR-04: Facilities Information

The system shall provide information about the major facilities and planned activities associated with the refinery project.

FR-05: Economic Benefits

The system shall display information about the expected economic benefits of the project, including industrial development, investment and business opportunities.

FR-06: Employment Opportunities

The system shall provide information about the expected employment and career opportunities associated with the project.

FR-07: Environmental Information

The system shall provide information about environmental and social aspects of the project, including available environmental assessments.

FR-08: Project Development Information

The system shall provide information about the progress and major development milestones of the project.

FR-09: Information Organization

The system shall organize information into different sections to make it easier for users to locate specific content.

3.2 News and Updates Management

FR-10: Display News

The system shall display news and announcements related to the refinery project.

FR-11: Add News

The system shall allow the administrator to add new news articles and announcements.

FR-12: Edit News

The system shall allow the administrator to edit existing news articles.

FR-13: Delete News

The system shall allow the administrator to remove outdated or incorrect news articles.

FR-14: Publish Updates

The system shall allow the administrator to publish important project updates and development milestones.

FR-15: Display Publication Dates

The system shall display the publication date of news articles and updates.

3.3 Public Inquiry Management

FR-16: Inquiry Form

The system shall provide an inquiry form that allows users to submit questions or comments.

FR-17: User Details

The inquiry form shall allow users to provide their names, email addresses and inquiry messages.

FR-18: Input Validation

The system shall validate information entered by users before accepting an inquiry.

FR-19: Store Inquiries

The system shall store submitted inquiries in the database.

FR-20: Administrator Access to Inquiries

The system shall allow the administrator to view inquiries submitted by users.

FR-21: Confirmation Message

The system shall display a confirmation message after an inquiry has been successfully submitted.

3.4 SMS Notification Management

FR-22: Subscriber Registration

The system shall allow users to register their phone numbers to receive important project announcements.

FR-23: Phone Number Validation

The system shall check that phone numbers entered by users follow the required format.

FR-24: Store Subscriber Information

The system shall store subscriber phone numbers and subscription information in the database.

FR-25: Send Notifications

The system shall support sending SMS notifications to registered users through an SMS gateway.

FR-26: Notification Management

The system shall allow the administrator to manage important notifications.

FR-27: Unsubscribe Option

The system shall provide a way for users to stop receiving SMS notifications.

FR-28: SMS Records

The system shall maintain records of SMS notifications sent through the system.

3.5 Administrator Management

FR-29: Administrator Login

The system shall provide a secure login page for authorized administrators.

FR-30: Login Validation

The system shall verify administrator login details before allowing access to the administration section.

FR-31: Unauthorized Access Prevention

The system shall prevent unauthorized users from accessing administrator functions.

FR-32: Dashboard

The system shall provide an administrator dashboard showing important website management information.

FR-33: Content Management

The system shall allow the administrator to add, edit, publish and delete website content.

FR-34: Inquiry Management

The system shall allow the administrator to view and manage inquiries submitted by website visitors.

FR-35: Subscriber Management

The system shall allow the administrator to view and manage SMS subscribers.

FR-36: Logout

The system shall allow administrators to securely log out of the system.

3.6 Search and Navigation

FR-37: Navigation Menu

The system shall provide a navigation menu that allows users to move between different website sections.

FR-38: Search Function

The system shall provide a search function that allows users to search for information available on the website.

FR-39: Search Results

The system shall display relevant results based on the search terms entered by the user.

FR-40: Image Gallery

The system shall provide a gallery for displaying relevant images related to the refinery project.

FR-41: External References

The system shall provide links to relevant official sources where users can obtain additional information.

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality standards and conditions that the system must meet.

4.1 Performance Requirements

NFR-01: The website should load pages within approximately three seconds under normal internet conditions.

NFR-02: The system should respond to user requests without unnecessary delays.

NFR-03: The database should retrieve stored information efficiently.

NFR-04: The system should support multiple users accessing the website at the same time.

4.2 Security Requirements

NFR-05: The system shall restrict administrator access to authorized users only.

NFR-06: Administrator passwords shall be stored using secure password hashing methods.

NFR-07: The system shall validate user input to reduce the risk of malicious data being submitted.

NFR-08: The system shall protect sensitive information from unauthorized access.

NFR-09: The system should use HTTPS to protect information transmitted between users and the website.

NFR-10: The system shall provide a secure logout function for administrators.

4.3 Usability Requirements

NFR-11: The website shall have a simple and user-friendly interface.

NFR-12: The system shall use clear menus and navigation links.

NFR-13: Information shall be arranged into clearly labelled sections.

NFR-14: Instructions and messages displayed to users shall be easy to understand.

NFR-15: Users should be able to access important information with minimal navigation.

4.4 Reliability Requirements

NFR-16: The system should operate consistently under normal operating conditions.

NFR-17: The system should display appropriate error messages when an operation fails.

NFR-18: The system should prevent incomplete or invalid information from being stored in the database.

NFR-19: The website should be available to users whenever the hosting service is operational.

4.5 Compatibility Requirements

NFR-20: The website shall be compatible with modern web browsers.

NFR-21: The system shall support access through desktop and laptop computers.

NFR-22: The system shall support access through smartphones and tablets.

4.6 Responsiveness Requirements

NFR-23: The website shall automatically adjust its layout according to the user's screen size.

NFR-24: Text, images, menus and other website elements shall remain readable on smaller screens.

NFR-25: The website shall provide a consistent user experience across different devices.

4.7 Maintainability Requirements

NFR-26: The system shall be organized into manageable sections to make future updates easier.

NFR-27: The administrator shall be able to update website information without changing the entire system.

NFR-28: The system should allow additional features to be introduced in the future.

4.8 Data Integrity and Backup Requirements

NFR-29: The system shall maintain accurate and consistent information in the database.

NFR-30: The system should prevent duplicate or incomplete records where possible.

NFR-31: Important website data should be backed up regularly.

NFR-32: The system should support the recovery of stored information from available backups.

5. SYSTEM USE CASES

A use case describes how a user interacts with the system to complete a particular activity.

5.1 Use Case: View Project Information

Actor: General Visitor

Description: A visitor accesses the website to learn about the refinery project.

Precondition: The website must be accessible.

Main Activities:

1. The visitor opens the website.
2. The system displays the homepage.
3. The visitor selects a section from the navigation menu.
4. The system displays the requested information.
5. The visitor reads the information.

Postcondition: The visitor successfully accesses the required project information.

5.2 Use Case: Submit an Inquiry

Actor: General Visitor

Description: A visitor submits a question or comment through the inquiry form.

Precondition: The visitor must have access to the inquiry form.

Main Activities:

1. The visitor opens the inquiry page.
2. The system displays the inquiry form.
3. The visitor enters the required details.
4. The system validates the information.
5. The visitor submits the form.
6. The system stores the inquiry in the database.
7. The system displays a confirmation message.

Postcondition: The inquiry is stored and made available to the administrator.

5.3 Use Case: Subscribe to SMS Notifications

Actor: General Visitor

Description: A visitor registers a phone number to receive project announcements.

Precondition: The subscription form must be available.

Main Activities:

1. The visitor opens the subscription section.
2. The system displays the subscription form.
3. The visitor enters a phone number.
4. The system validates the phone number.
5. The visitor submits the form.
6. The system stores the subscriber information.
7. The system displays a confirmation message.

Postcondition: The visitor is registered as a subscriber.

5.4 Use Case: Administrator Login

Actor: Administrator

Description: The administrator logs into the system to manage website content.

Precondition: The administrator must have valid login details.

Main Activities:

1. The administrator opens the login page.
2. The system displays the login form.
3. The administrator enters a username and password.
4. The system verifies the login details.
5. The system grants access if the details are correct.
6. The administrator is directed to the dashboard.

Postcondition: The administrator gains access to authorized management functions.

5.5 Use Case: Manage Website Content

Actor: Administrator

Description: The administrator updates information displayed on the website.

Precondition: The administrator must be logged in.

Main Activities:

1. The administrator opens the dashboard.
2. The administrator selects the content management section.
3. The administrator adds, edits or deletes information.
4. The system processes the changes.
5. The system stores the updated information.
6. The updated content becomes available on the public website.

Postcondition: Website information is updated.

5.6 Use Case: Publish News and Announcements

Actor: Administrator

Description: The administrator publishes news and announcements about the project.

Precondition: The administrator must be logged in.

Main Activities:

1. The administrator opens the news management section.
2. The administrator enters the news title and content.
3. The administrator adds relevant images or documents where necessary.
4. The administrator submits the information.
5. The system stores and publishes the news.
6. Visitors can view the published announcement.

Postcondition: The news article is available on the website.

5.7 Use Case: Send SMS Notifications

Actor: Administrator

Description: The administrator sends important announcements to registered subscribers.

Precondition: Subscribers must be registered and an SMS gateway must be available.

Main Activities:

1. The administrator opens the notification section.
2. The administrator enters the announcement.
3. The administrator selects the intended subscribers.
4. The system sends the message through the SMS gateway.
5. The system records the notification results.

Postcondition: The system records the SMS notification activity.

6. DATABASE REQUIREMENTS

6.1 Database Description

The system will use a database to store and manage information required by the website.

The database will store administrator details, project articles, news updates, inquiries, subscriber information and SMS notification records.

MySQL is proposed as the database management system.

6.2 Database Tables

The database will contain the following main tables:

1. Users Table

This table will store administrator account information.

Fields include:

- User ID
- Username
- Email
- Password Hash
- Date Created

2. Articles Table

This table will store general information about the refinery project.

Fields include:

- Article ID

- Article Title
- Article Content
- Category ID
- Image
- Date Created
- Date Updated

3. News Updates Table

This table will store news articles and project announcements.

Fields include:

- News ID
- News Title
- News Content
- Image
- Publication Date
- Administrator ID

4. Inquiries Table

This table will store inquiries submitted by website visitors.

Fields include:

- Inquiry ID
- Visitor Name
- Email Address
- Message
- Date Submitted
- Status

5. Subscribers Table

This table will store information about users who subscribe to SMS notifications.

Fields include:

- Subscriber ID
- Phone Number
- Subscription Status
- Date Registered

6. SMS Logs Table

This table will store records of SMS notifications sent through the system.

Fields include:

- SMS ID
- Subscriber ID
- Message
- Date Sent
- Delivery Status

7. Categories Table

This table will organize project information into different categories.

Fields include:

- Category ID
- Category Name
- Category Description

6.3 Database Relationships

The database will have the following relationships:

1. One administrator can publish many news articles.
2. One category can contain many project articles.
3. One subscriber can receive many SMS notifications.
4. The inquiry table will store messages submitted by different website visitors.
5. The articles table will be linked to the categories table through the Category ID.

These relationships will help maintain organized and consistent information within the database.

6.4 Data Integrity Requirements

The system shall ensure that:

1. Each database record has a unique identification number.
2. Required fields are completed before information is saved.
3. Email addresses and phone numbers are validated.
4. Related records use valid identification numbers.
5. Unauthorized users cannot modify protected records.
6. Duplicate subscriber records are prevented where possible.

7. BUSINESS RULES

The system shall operate according to the following business rules:

1. Only authorized administrators shall be allowed to access the administration section.
2. Visitors shall be allowed to view publicly available information without logging in.
3. Only administrators shall be allowed to publish or edit official website content.
4. All published information should be checked before being made available to the public.
5. Users must provide valid information when submitting inquiries.
6. Users must provide a valid phone number when subscribing to SMS notifications.
7. Only registered subscribers shall receive SMS notifications.
8. Subscribers shall be allowed to unsubscribe from notifications.
9. Every published news article shall have a publication date.
10. The system shall store submitted inquiries for administrator access.
11. Administrators shall be responsible for keeping website information updated.
12. Information displayed on the website should be obtained from reliable sources.
13. The system shall prevent unauthorized access to administrator functions.

8. SYSTEM CONSTRAINTS AND ASSUMPTIONS

8.1 System Constraints

The system will have the following limitations:

1. Users must have an internet connection to access the website.
2. The website will depend on a hosting service to remain available online.
3. SMS notifications will depend on an external SMS gateway.
4. The system will only provide information available from accessible and reliable sources.
5. The system will not control or monitor the actual industrial operations of the refinery.
6. The system will not process payments or manage petroleum sales.
7. The administrator will be responsible for maintaining and updating website information.

8.2 System Assumptions

The development of the system will be based on the following assumptions:

1. Users will have access to internet-enabled devices.
2. Most users will access the website using modern browsers.
3. The administrator will have the necessary login credentials.
4. Reliable information about the refinery project will be available from official sources and publications.
5. The website will be hosted on a suitable web server.
6. The database will be available when information needs to be stored or retrieved.
7. Users who subscribe to notifications will provide correct contact information.

9. SYSTEM TESTING AND ACCEPTANCE CRITERIA

9.1 Testing Requirements

The system shall be tested to confirm that it meets the specified requirements and performs its intended functions.

The following areas shall be tested:

1. Functional Testing

This will confirm that the website functions correctly, including displaying information, submitting inquiries, subscribing to notifications and managing content.

2. Usability Testing

This will assess whether users can navigate the website easily and understand the information displayed.

3. Compatibility Testing

This will confirm that the website works correctly on different browsers and devices.

4. Security Testing

This will check whether administrator access is protected and user input is properly validated.

5. Performance Testing

This will assess how quickly the website loads and responds to user requests under normal conditions.

9.2 Acceptance Test Cases

Test Case 1: Homepage Display

- Action: Open the website.
- Expected Result: The homepage should display correctly.

Test Case 2: Navigation

- Action: Select different sections from the navigation menu.
- Expected Result: The system should open the selected pages.

Test Case 3: Project Information

- Action: Open the project background section.
- Expected Result: The system should display the available project information.

Test Case 4: News Display

- Action: Open the news section.

- Expected Result: Published news articles should be displayed.

Test Case 5: Inquiry Submission

- Action: Complete and submit the inquiry form.
- Expected Result: The inquiry should be stored and a confirmation message displayed.

Test Case 6: Invalid Inquiry Details

- Action: Submit an inquiry with missing or invalid required information.
- Expected Result: The system should display an appropriate validation message.

Test Case 7: Administrator Login

- Action: Enter correct administrator login details.
- Expected Result: The administrator should access the dashboard.

Test Case 8: Invalid Administrator Login

- Action: Enter incorrect login details.
- Expected Result: The system should deny access and display an error message.

Test Case 9: Content Management

- Action: Add or edit a website article as an administrator.
- Expected Result: The system should save the changes correctly.

Test Case 10: News Publication

- Action: Publish a news article.
- Expected Result: The article should appear in the public news section.

Test Case 11: SMS Subscription

- Action: Enter a valid phone number and submit the subscription form.
- Expected Result: The subscriber details should be stored.

Test Case 12: Invalid Phone Number

- Action: Enter an invalid phone number.
- Expected Result: The system should display a validation message.

Test Case 13: SMS Notification

- Action: Send a notification to registered subscribers.
- Expected Result: The system should submit the message through the SMS gateway and record the result.

Test Case 14: Mobile Responsiveness

- Action: Open the website using a smartphone.
- Expected Result: The website should adjust to the screen size and remain usable.

Test Case 15: Search Function

- Action: Search for a topic available on the website.
- Expected Result: The system should display relevant information.

9.3 Acceptance Criteria

The system will be considered acceptable when:

1. Users can access the website successfully.
2. All main website sections display the correct information.
3. Users can navigate between different sections.
4. The inquiry form accepts valid submissions.
5. The administrator can log in securely.
6. The administrator can manage website content.
7. The system can store subscriber information.
8. SMS notifications can be processed when the SMS gateway is available.
9. The website works on different screen sizes.
10. The system meets the stated functional and non-functional requirements.

10. FUTURE ENHANCEMENTS

The system may be improved in the future by introducing additional features such as:

1. Multiple Language Support: The website may support additional languages to reach more users.

2. Interactive Maps: Maps may be introduced to help users understand the geographical location of the refinery.
3. Advanced Search: The search function may be improved to provide more accurate results.
4. Email Notifications: Users may receive important announcements through email.
5. Social Media Integration: The website may include links to relevant social media platforms.
6. Downloadable Documents: Users may be allowed to download official reports and publications.
7. Website Analytics: The system may include tools for monitoring website visits and user activity.
8. Accessibility Improvements: Additional features may be introduced to make the website easier to use for people with different accessibility needs.

11. CONCLUSION

This Software Requirements Specification describes the requirements of the proposed Dangote East Africa Petroleum Refinery Informative Website.

The system is intended to provide a centralized platform where users can access information about the refinery project, including its background, location, facilities, expected economic benefits, environmental aspects and development progress.

The website will also provide features such as news updates, public inquiries, SMS notifications and administrator content management.

By bringing information together in one organized platform, the system will make it easier for students, researchers and members of the public to access information about the project. The requirements outlined in this document will guide the development of a simple, responsive and user-friendly website.

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